Algebra 1 – Module 3 Test PRACTICE **WORKED KEY**

Module 3 — Relations and Functions

Answer each question, making sure to show your work in the space provided. Take the time to check your work. Write neatly. Circle your final answer.

1. Vocabulary. Write the letter of the term that matches each description. One term is never used. (4)

A) function
B) domain
C) relative minimum
D) continuous function
E) x-intercept

a. **B** the set of all x-values (inputs) of a relation

b. **E** the point where a graph crosses the x-axis

c. **C** a point that is lower than all the nearby points on a graph

d. **D** a function that can be graphed with a line or a smooth curve

2. State the domain and range of the relation {(1, −6), (−5, 2), (3, 2), (0, 9)}. (2)

→ D = {−5, 0, 1, 3}; R = {−6, 2, 9}

3. Identify the independent and dependent variables: "The more miles you drive, the more gas you use." (2)

→ independent: miles driven; dependent: gas used

4. Is the function discrete or continuous? Explain: the number of cars in a parking lot, counted each hour. (2)

→ discrete: you can only count whole cars, so the graph is separate points

5. Is the relation a function? Explain. {(3, 1), (−2, 4), (3, 6), (0, −1)} (2)

→ NOT A FUNCTION: x = 3 is paired with two different y-values (1 and 6)

6. $f(x) = 5x - 3$. Find $f(2) + f(-1)$. (2)

→ $f(2) = 7$, $f(-1) = -8$ → $7 + (-8) = -1$

7. Sam earns $g(x) = 10.25x$ dollars for babysitting x hours. Find $g(4)$. Then explain what it means. (2)

→ $g(4) = 10.25(4) = 41.00$ → Sam earns $41.00 for 4 hours

8. Is each function linear or nonlinear? Explain how you know. (2)

a) $y = 3x + 2x - 8$

b) $y = x^2 + 4x$

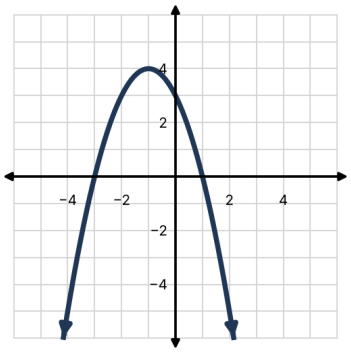
→ a) linear (simplifies to $y = 5x - 8$); b) nonlinear (x is squared)

9. Find the x- and y-intercepts of $4x - 3y = 12$. (2)

→ x-intercept: set $y = 0$ → $x = 3$ → (3, 0); y-intercept: set $x = 0$ → $y = -4$ → (0, −4)

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Use the graph of f(x) for questions 10 – 11.



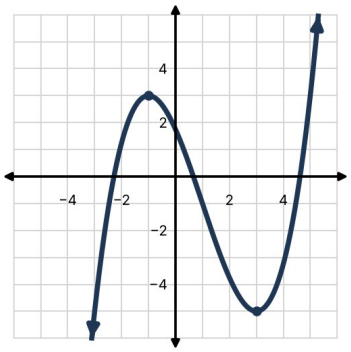
10. Name the x- and y-intercepts. Then describe where f(x) is positive and where it is negative. (2)

→ x-intercepts (−3, 0) and (1, 0); y-intercept (0, 3); positive: −3 < x < 1; negative: x < −3 and x > 1

11. Name the line of symmetry. Then describe the end behavior. (2)

→ line of symmetry x = −1; end behavior: as x increases or decreases, y decreases

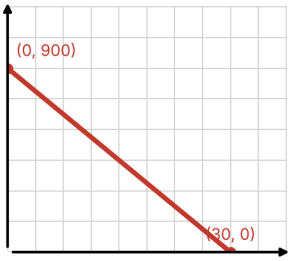
Use the graph of g(x) for question 12.



12. Over which intervals is g(x) increasing? Decreasing? Name every relative maximum and relative minimum. (2)

→ increasing: x < −1 and x > 3; decreasing: −1 < x < 3; rel. max (−1, 3), rel. min (3, −5)

13. A 900-gallon tank drains at a steady rate. It is empty after 30 minutes. Sketch a graph of gallons left (y) vs. minutes (x). Label both intercepts and tell what each one means. (2)



→ line from (0, 900) down to (30, 0): y-intercept (0, 900) = the starting amount; x-intercept (30, 0) = when it runs out; linear and decreasing

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Cumulative Review — ACT Style. Circle the best answer. No work required, but scratch work never hurts.

14. Solve: 2(x − 5) = x + 4 (1)

A) 9
B) 14
C) −6
D) 1

→ B 2x − 10 = x + 4 → x = 14

15. Which is equivalent to 5(x − 2) + 3x ? (1)

E) 8x − 10
F) 8x − 2
G) 2x − 10
H) 8x + 10

→ E 5x − 10 + 3x = 8x − 10